📌 Problem Statement

In the modern education system, a huge amount of student-related data is generated every day, including marks, subjects, attendance, and performance records. Managing and analyzing this large volume of data using traditional methods such as manual calculations or simple software tools becomes difficult, time-consuming, and inefficient.

As the size of the dataset increases, it becomes challenging to process the data quickly and accurately. Educational institutions require efficient systems to analyze student performance in order to identify trends, evaluate results, and make data-driven decisions. Traditional data processing techniques are not capable of handling large-scale datasets efficiently, especially when real-time or fast processing is required.

One of the major challenges is to extract useful insights from raw data, such as calculating total marks, average marks, identifying top-performing students, and performing subject-wise analysis. Without proper tools, these operations require a lot of time and effort, and there is a higher chance of errors.

To overcome these challenges, there is a need for a scalable, reliable, and efficient data processing system that can handle large datasets and perform complex computations quickly. Big data technologies such as Hadoop provide a distributed environment where data can be processed across multiple systems, improving speed and efficiency.

This project focuses on solving these problems by using Hadoop MapReduce and Hive. Hadoop MapReduce allows distributed processing of large datasets by dividing the data into smaller parts and processing them in parallel. Hive provides a SQL-like interface to query and analyze the data easily.

By using these technologies, this project aims to efficiently analyze student performance data and generate meaningful insights such as total marks, average marks, maximum and minimum marks, and the count of students per subject. This approach not only improves processing speed but also ensures accuracy and scalability.

This problem is highly relevant in real-world scenarios where organizations deal with large amounts of data and require efficient tools for analysis. Hence, implementing a big data solution for student performance analysis helps in understanding the practical use of Hadoop and its importance in modern data processing systems.